

STAPLES, MN, USA

WMO: 727514

Lat: 46.381N

Lon: 94.806W

Elev: 1287

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 54910

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.0	-15.4	-26.2	1.3	-19.2	-19.9	1.9	-14.3	25.4	16.8	22.8	16.1	4.3	270	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	88.0	74.3	84.0	72.2	81.7	69.8	77.5	84.5	75.0	81.5	72.8	78.7	10.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	138.2	81.7	72.8	127.7	79.0	71.6	122.5	77.6	41.9	84.3	39.6	81.7	37.2	80.0	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.4	19.2	17.2	DB	-27.6	92.2	6.6	3.1	-32.3	94.4	-36.1	96.2	-39.8	97.9	-44.6	100.1	(4)
			WB	-27.8	80.2	6.5	3.2	-32.5	82.5	-36.3	84.4	-39.9	86.1	-44.6	88.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	10.5	12.9	27.5	41.9	54.5	64.3	69.6	66.6	58.5	44.3	30.5	15.6	(6)
(7)		DBStd	23.15	14.04	13.12	13.30	10.35	8.54	6.46	5.70	5.68	8.50	8.86	11.23	12.81	(7)
(8)		HDD50	5195	1225	1040	704	278	48	1	0	0	20	222	589	1068	(8)
(9)		HDD65	8947	1690	1460	1164	693	342	88	22	48	229	643	1035	1533	(9)
(10)		CDD50	2104	0	0	5	36	188	429	607	514	275	46	4	0	(10)
(11)		CDD65	380	0	0	0	1	17	66	164	97	34	1	0	0	(11)
(12)		CDH74	3727	0	0	0	19	235	662	1569	906	313	23	0	0	(12)
(13)		CDH80	988	0	0	0	3	64	148	466	235	68	4	0	0	(13)
(14)	Wind	WSAvg	7.0	7.2	7.2	7.4	7.9	7.7	6.6	5.7	5.3	6.5	7.3	7.5	7.1	(14)
(15)	Precipitation	PrecAvg	27.4	0.7	0.6	1.4	2.2	3.5	4.3	4.0	3.1	2.8	2.6	1.3	0.8	(15)
(16)		PrecMax	34.4	2.1	1.7	3.5	5.4	7.0	7.1	9.0	6.9	6.9	5.9	4.1	1.9	(16)
(17)		PrecMin	20.1	0.0	0.1	0.4	0.9	1.1	1.5	1.6	0.9	0.7	0.4	0.2	0.3	(17)
(18)		PrecStd	3.5	0.5	0.4	0.7	1.0	1.5	1.5	1.7	1.5	1.3	1.6	1.0	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	46.3	65.7	75.5	87.8	88.4	91.5	90.3	86.3	78.7	64.0	43.5	(19)
(20)		0.4%	MCWB	38.5	42.6	56.0	60.4	67.8	74.3	77.9	75.9	71.4	62.3	54.8	39.2	(20)
(21)		2%	DB	36.2	40.6	57.2	70.0	81.0	83.8	88.2	85.6	81.5	70.2	56.9	37.4	(21)
(22)		2%	MCWB	34.0	37.0	49.0	55.8	64.2	72.6	75.7	72.7	69.4	59.0	49.7	35.4	(22)
(23)		5%	DB	33.5	36.3	51.7	64.3	74.8	81.1	84.2	82.0	78.6	63.7	51.9	35.6	(23)
(24)		5%	MCWB	32.5	34.2	47.0	52.0	61.2	68.5	73.4	70.5	67.4	55.1	46.0	34.3	(24)
(25)		10%	DB	29.6	32.0	45.1	60.8	71.5	78.6	81.9	79.3	73.0	59.1	45.9	32.1	(25)
(26)		10%	MCWB	29.2	31.1	41.1	49.4	59.6	66.6	71.0	68.7	65.0	52.0	42.4	31.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	43.4	57.7	63.9	72.7	78.2	81.9	80.2	75.7	66.7	57.8	40.2	(27)
(28)		0.4%	MCDWB	42.5	46.8	63.3	75.2	80.1	85.0	87.6	84.9	83.2	72.3	61.4	42.3	(28)
(29)		2%	WB	34.2	37.7	51.5	57.7	67.3	75.0	78.6	75.5	72.7	60.8	50.7	36.4	(29)
(30)		2%	MCDWB	35.5	40.1	54.3	66.2	75.9	81.3	85.4	82.0	78.3	68.0	56.8	38.1	(30)
(31)		5%	WB	31.8	34.2	45.8	53.8	63.7	71.9	75.8	73.2	69.7	56.6	46.1	34.0	(31)
(32)		5%	MCDWB	32.9	35.8	50.1	62.7	70.6	77.5	82.3	79.2	75.0	62.6	49.7	35.1	(32)
(33)		10%	WB	28.4	30.4	41.2	50.3	60.6	68.5	73.4	70.9	66.6	53.2	42.4	31.0	(33)
(34)		10%	MCDWB	28.8	31.8	45.2	59.3	68.4	75.2	79.4	76.6	71.8	58.0	46.4	31.9	(34)
(35)	Mean Daily Temperature Range	MDBR	18.0	19.7	19.0	20.7	21.5	20.5	21.4	21.4	20.9	18.1	16.2	15.4	(35)	
(36)		5% DB	MCDBR	17.6	20.1	25.3	29.5	29.5	24.8	24.2	24.9	25.9	26.1	24.6	17.8	(36)
(37)		5% DB	MCWBR	15.3	16.4	17.2	17.4	15.8	13.1	12.9	13.2	14.6	16.1	16.6	14.9	(37)
(38)		5% WB	MCDWR	17.1	18.5	23.4	26.5	24.5	20.8	22.0	21.6	21.3	23.4	22.3	16.6	(38)
(39)		5% WB	MCWBR	15.0	15.5	16.6	16.9	15.4	13.8	13.6	13.4	13.9	16.0	16.2	14.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.260	0.294	0.307	0.358	0.379	0.379	0.381	0.385	0.354	0.311	0.282	0.260	(40)	
(41)		taud	2.374	2.276	2.381	2.324	2.328	2.362	2.371	2.370	2.437	2.541	2.511	2.411	(41)	
(42)		Ebn at Noon	270	279	293	285	281	281	279	273	273	272	261	257	(42)	
(43)		Edh at Noon	23	31	32	38	39	38	37	36	30	24	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	514	868	1214	1533	1705	1872	1990	1673	1254	777	482	355	(44)	
(45)		RadStd	60	93	119	127	126	124	83	144	94	119	49	46	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	-4.66	-5.00	N/A	-4.81	-5.15	N/A	N/A	N/A	N/A	
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page